Lab Assignment 5

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General Context

The Advanced Adversary Attack Tactics Cyber Range, specifically Reconnaissance and Resource Development and Pivoting with Proxychains, focused on initial access preparation and post-compromise lateral movement. The first lab focused on reconnaissance as a structured intelligence gathering process, where, rather than exploiting technical vulnerability immediately. It showed how attackers collect publicly available information to build credibility and knowledge before launching a phishing or credential harvesting campaign. This is a representation or real world adversary behavior, which is part of the cyber kill chain model, where reconnaissance and resource development are significant in influencing successful attacks. Industry reports continue to show that social engineering and credential compromise remain dominant breach vectors, reinforcing the importance of understanding how attackers weaponize publicly accessible information (IBM Security 2023).

The Pivoting with Proxychains lab focused more on the post compromise, specifically network pivoting and traffic tunneling. It showed how an attacker, once they gain access to one system, can leverage it as an intermediary to reach other protected internal assets. It directly covers lateral movement and command-and-control techniques outlined in the MITRE ATT&CK framework (MITRE, 2023). It emphasises how perimeter security alone is not enough, and organizations must include segmentation and detection for anomalous proxy behavior. The lab also showed how encrypted SSH tunnels and SOCKS proxies can obscure the true origin of traffic, making malicious activity appear as though it is coming from a trusted internal host.

In the Reconnaissance and Resource Development lab, **cewl -n -e --email_file=./emails** http://target/login was used to collect email addresses, and then **gobuster dir -w /usr/share/seclists/Discovery/Web-Content/common.txt -u target/ -x "txt,md"** found hidden directories such as /downloads and /news. Then, using the Social Engineering Toolkit, cloned the login page to capture submitted credentials, which simulated the credential harvesting. The Pivoting with Proxychains lab had us use **nmap -p- pivot** to identify exposed services, followed by **ssh -f -N -D 9050 root@pivot -4** to establish a SOCKS proxy tunnel. After configuring ProxyChains to use 127.0.0.1 9050, the commands **proxychains4 nmap -Pn -sT target** and **proxychains4 firefox** were run to access otherwise unreachable internal systems, illustrating practical lateral movement and tunneling techniques.

Technical Context

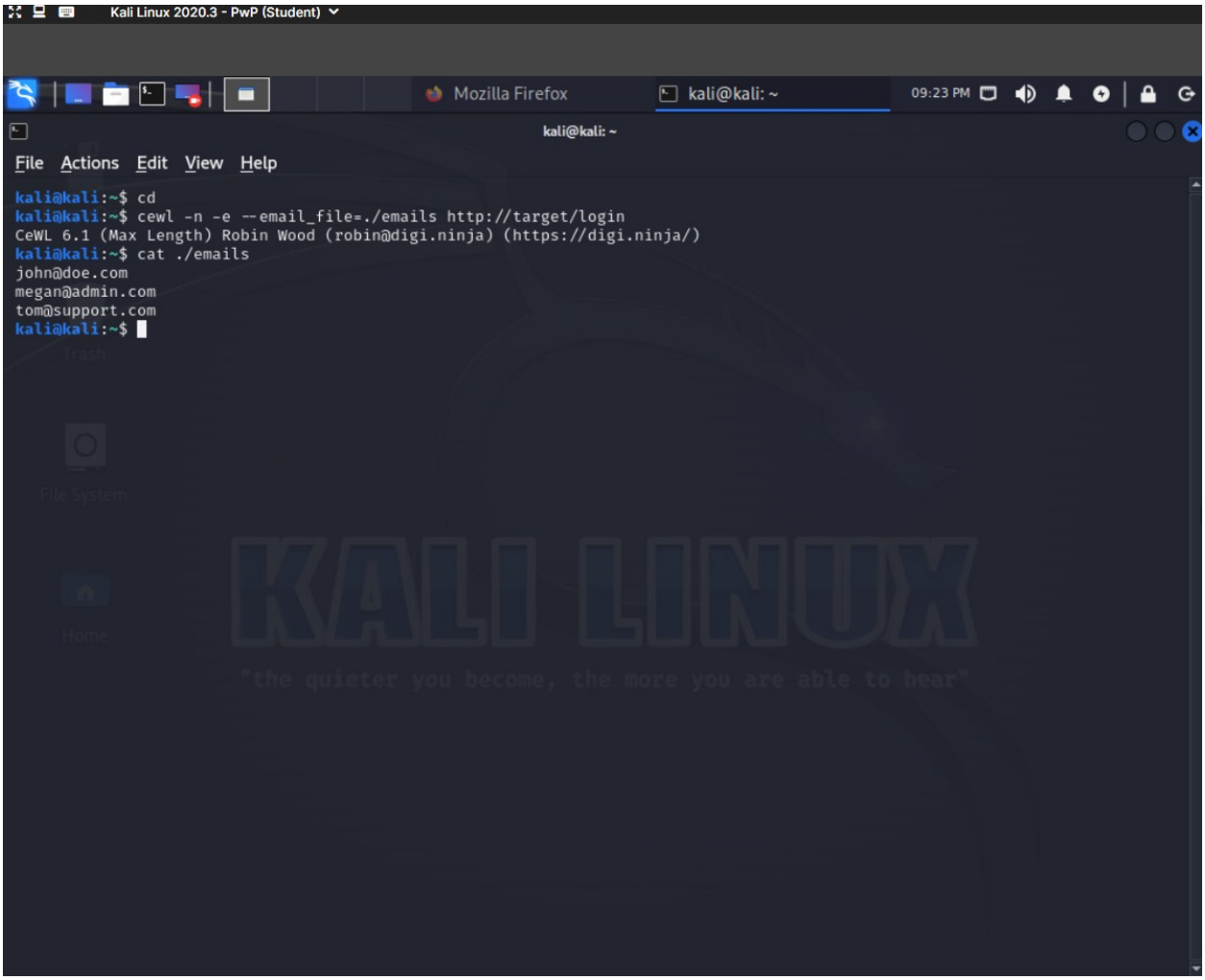
Reconnaissance and Resource Development and Pivoting with Proxychains, focused on initial access preparation and post-compromise lateral movement. The first lab centered on reconnaissance as a structured intelligence gathering process, where, rather than exploiting a technical vulnerability immediately, the objective was to collect publicly available information to build credibility and operational knowledge before launching a phishing or credential harvesting campaign. Commands such as **cewl -n -e --email_file=./emails http://target/login** were used to extract email addresses directly from the target site, and **gobuster dir -w /usr/share/seclists/Discovery/Web-Content/common.txt -u target/ -x "txt,md"** identified hidden directories like /downloads and /news. This represents real-world adversary behavior consistent with the cyber kill chain model, where reconnaissance and resource development significantly influence the likelihood of a successful attack. Industry research continues to show that phishing and compromised credentials remain dominant initial access vectors, reinforcing how attackers weaponize exposed information (IBM Security, 2023), and these risks align with security misconfiguration issues identified in the OWASP Top Ten (OWASP Foundation, 2021).

The Pivoting with Proxychains lab focused more on the post compromise phase, specifically network pivoting and traffic tunneling. It demonstrated how, once access to a single system is obtained, that system can be leveraged as an intermediary to reach additional protected internal assets. The lab used **nmap -p- pivot** to enumerate services, followed by **ssh -f -N -D 9050 root@pivot -4** to establish a dynamic SOCKS proxy tunnel. After configuring ProxyChains to use 127.0.0.1 9050, commands such as **proxychains4 nmap -Pn -sT target** and **proxychains4 firefox** allowed access to systems that were not directly reachable. This directly reflects lateral movement and encrypted command-and-control techniques outlined in the MITRE ATT&CK framework (MITRE, 2023). It emphasizes that perimeter security alone is insufficient, and organizations must implement internal segmentation and monitoring controls.

The two labs overall covered reconnaissance, credential harvesting, SSH tunneling, and proxy-based routing, showing how it can all be chained together into an adversary attack workflow. From collecting emails and cloning a login page with SET to pivoting traffic through encrypted tunnels, we saw how limited initial access can expand into broader internal compromise. Federal guidance from NIST highlights the importance of firewall policy and internal network controls to limit such movement (NIST, 2009), and CISA guidance further stresses detection of lateral movement and anomalous proxy behavior within enterprise networks (CISA, 2022). Together, these labs reinforce the need for layered defensive strategies that address both exposed public information and internal network visibility.

Screenshots

Reconnaissance and Resource Development



Exploring Hidden Directories for Valuable Information

```
kali@kali:~$ gobuster dir -w /usr/share/seclists/Discovery/Web-Content/common.txt -u target/ -x "txt,md" | grep -v "Size: 2
71"
===============================================================
Gobuster v3.6
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://target/
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/seclists/Discovery/Web-Content/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.6
[+] Extensions:              md,txt
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
/downloads            (Status: 301) [Size: 304] [--> http://target/downloads/]
/index.html           (Status: 200) [Size: 19]
/javascript           (Status: 301) [Size: 305] [--> http://target/javascript/]
/login                (Status: 301) [Size: 300] [--> http://target/login/]
/news                 (Status: 301) [Size: 299] [--> http://target/news/]
Progress: 14202 / 14205 (99.98%)
===============================================================
Finished
===============================================================
kali@kali:~$
```

Exploring Hidden Directories for Valuable Information

```
Kali Linux 2020.3 - PwP (Student)
Mozilla Firefox    kali@kali: ~    09:23 PM
kali@kali: ~
File  Actions  Edit  View  Help
CeWL 6.1 (Max Length) Robin Wood (robin@digi.ninja) (https://digi.ninja/)
kali@kali:~$ cat ./emails
john@doe.com
megan@admin.com
tom@support.com
kali@kali:~$ gobuster dir -w /usr/share/seclists/Discovery/Web-Content/common.txt -u target/ -x "txt,md"
===============================================================
Gobuster v3.6
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://target/
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/seclists/Discovery/Web-Content/common.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.6
[+] Extensions:              txt,md
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
/.hta                 (Status: 403) [Size: 271]
/.hta.txt             (Status: 403) [Size: 271]
/.hta.md              (Status: 403) [Size: 271]
/.htaccess.txt        (Status: 403) [Size: 271]
/.htaccess.md         (Status: 403) [Size: 271]
/.htaccess            (Status: 403) [Size: 271]
/.htpasswd            (Status: 403) [Size: 271]
/.htpasswd.txt        (Status: 403) [Size: 271]
/.htpasswd.md         (Status: 403) [Size: 271]
/downloads            (Status: 301) [Size: 304] [--> http://target/downloads/]
/index.html           (Status: 200) [Size: 19]
/javascript           (Status: 301) [Size: 305] [--> http://target/javascript/]
/login                (Status: 301) [Size: 300] [--> http://target/login/]
/news                 (Status: 301) [Size: 299] [--> http://target/news/]
/server-status        (Status: 403) [Size: 271]
Progress: 14202 / 14205 (99.98%)
===============================================================
Finished
===============================================================
kali@kali:~$
```

Cloning a Website for Credential Harvesting with SET

Conclusion - Post clicking Login button on http://localhost page

```
kali@kali: ~
File  Actions  Edit  View  Help
   2) Site Cloner
   3) Custom Import

  99) Return to Webattack Menu

set:webattack>2
[-] Credential harvester will allow you to utilize the clone capabilities within SET
[-] to harvest credentials or parameters from a website as well as place them into a report

─────────────────────────────────────────────────────────────────────────────
── * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * ──

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesns't know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perpective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.1.100]:target
[-] SET supports both HTTP and HTTPS
[-] Example: http://www.thisisafakesite.com
set:webattack> Enter the url to clone:http://target/login

[*] Cloning the website: http://target/login
[*] This could take a little bit...

The best way to use this attack is if username and password form fields are available. Regardless, this captures all POSTs
on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
```

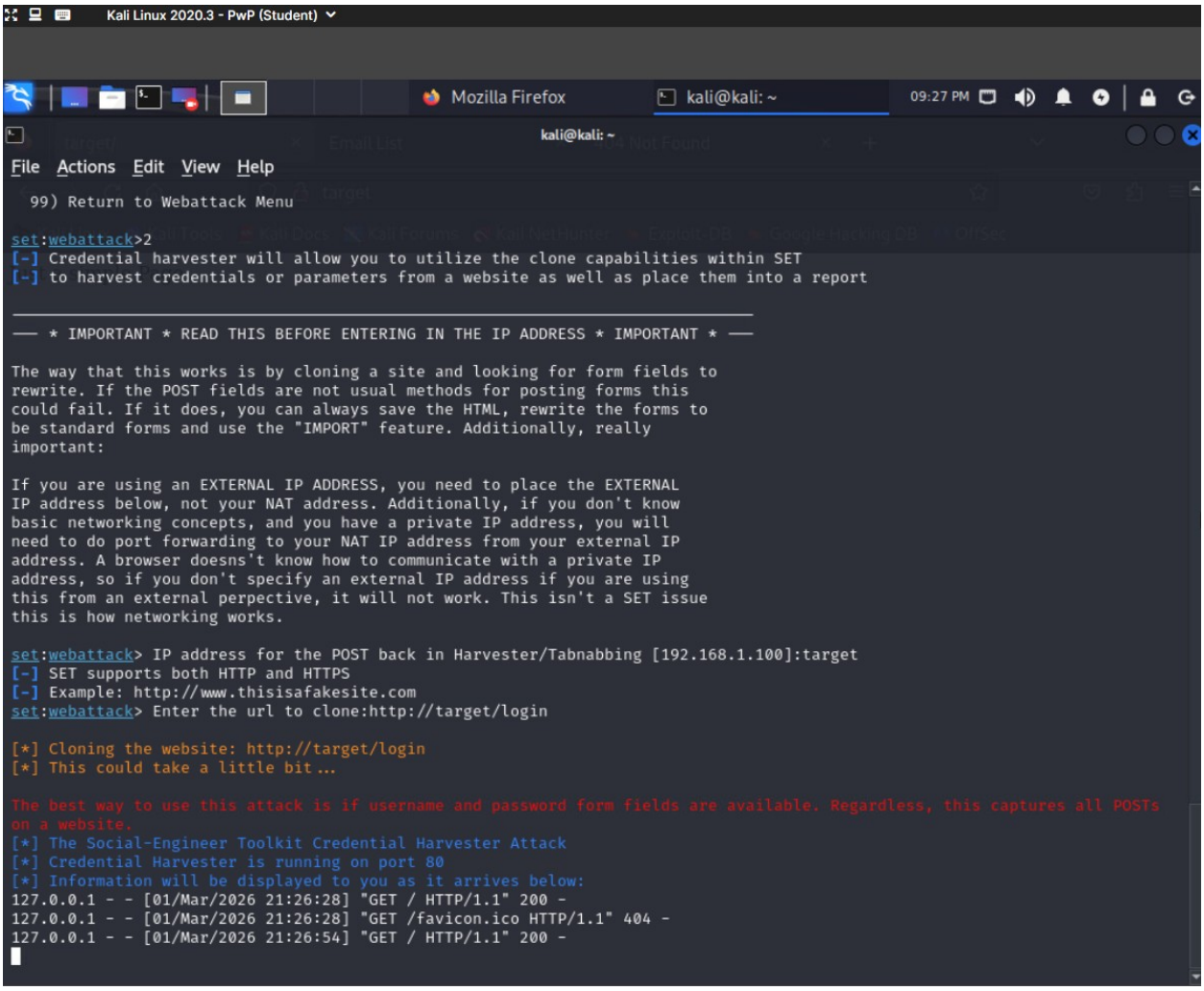
```
5000/tcp open  upnp

Nmap done: 1 IP address (1 host up) scanned in 1.07 seconds
kali@kali:~$ ssh root@pivot
The authenticity of host 'pivot (192.168.1.101)' can't be established.
ED25519 key fingerprint is SHA256:8lAc2qAXsTlzjsrQbQhbDrc8dnNtVcG35gkI2YXrcgs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'pivot' (ED25519) to the list of known hosts.
root@pivot's password:
Permission denied, please try again.
root@pivot's password:
Linux kali 6.5.0-kali3-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.5.6-1kali1 (2023-10-09) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
┌──(Message from Kali developers)
│
│ We have kept /usr/bin/python pointing to Python 2 for backwards
│ compatibility. Learn how to change this and avoid this message:
│ ⇒ https://www.kali.org/docs/general-use/python3-transition/
│
└──(Run: "touch ~/.hushlogin" to hide this message)
root@kali:~# nmap -p- target
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-01 21:32 EST
Nmap scan report for target (192.168.1.102)
Host is up (0.000046s latency).
Not shown: 65531 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
3310/tcp open  dyna-access
3389/tcp open  ms-wbt-server
MAC Address: 00:0C:29:10:95:E8 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.91 seconds
root@kali:~#
```



Pivoting with Proxychains

```
kali@kali: ~
File  Actions  Edit  View  Help
5000/tcp open  upnp

Nmap done: 1 IP address (1 host up) scanned in 1.07 seconds
kali@kali:~$ ssh root@pivot
The authenticity of host 'pivot (192.168.1.101)' can't be established.
ED25519 key fingerprint is SHA256:8lAc2qAXsTlzjsrQbQhbDrc8dnNtVcG35gkI2YXrcgs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'pivot' (ED25519) to the list of known hosts.
root@pivot's password:
Permission denied, please try again.
root@pivot's password:
Linux kali 6.5.0-kali3-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.5.6-1kali1 (2023-10-09) x86_64

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┌──(Message from Kali developers)
│
│ We have kept /usr/bin/python pointing to Python 2 for backwards
│ compatibility. Learn how to change this and avoid this message:
│ ⇒ https://www.kali.org/docs/general-use/python3-transition/
│
└──(Run: "touch ~/.hushlogin" to hide this message)
root@kali:~# nmap -p- target
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-01 21:32 EST
Nmap scan report for target (192.168.1.102)
Host is up (0.000046s latency).
Not shown: 65531 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
3310/tcp open  dyna-access
3389/tcp open  ms-wbt-server
MAC Address: 00:0C:29:10:95:E8 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.91 seconds
root@kali:~#
```

Setting Up Proxychains for Target Interaction

```
kali@kali: ~
File  Actions  Edit  View  Help
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1100 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:38292 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:3371 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1059 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:4129 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:7676 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1069 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:2009 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:7106 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1148 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1063 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:49153 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:667 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1271 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:119 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1082 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:32 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:8022 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:33354 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:55555 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:8701 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:593 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:3784 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1310 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:7007 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:5915 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:1106 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:2381 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:7496 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:5120 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:19350 <--socket error or timeout!
Nmap scan report for target (192.168.1.102)
Host is up (0.00037s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
3389/tcp open  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
kali@kali:~$
```

Using Proxychains with Multiple Layers

```
File  Actions  Edit  View  Help
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  contile.services.mozilla.com:443 [proxychains] DLL init: proxychains-
ng 4.16
 ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  spocs.getpocket.com:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  content-signature-2.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  push.services.mozilla.com:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  safebrowsing.googleapis.com:443  ...  OK
[proxychains] DLL init: proxychains-ng 4.16
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  firefox.settings.services.mozilla.com:443 [proxychains] DLL init: pro
xychains-ng 4.16
[proxychains] DLL init: proxychains-ng 4.16
 ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  o.pki.goog:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  push.services.mozilla.com:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  firefox.settings.services.mozilla.com:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  firefox-settings-attachments.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  ads-img.mozilla.org:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  ads-img.mozilla.org:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  192.168.1.102:80  ...  OK
[proxychains] DLL init: proxychains-ng 4.16
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  shavar.services.mozilla.com:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  tracking-protection.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  normandy.cdn.mozilla.net:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  services.addons.mozilla.org:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  location.services.mozilla.com:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  aus5.mozilla.org:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  aus5.mozilla.org:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  classify-client.services.mozilla.com:443  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  ciscobinary.openh264.org:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  archive.mozilla.org:443  ...  OK
[proxychains] DLL init: proxychains-ng 4.16
```

Lab Completions

Advanced Adversary Tactics - Reconnaissance and Resource Development
Lab · 1 hour

The first steps of the MITRE ATT&CK chain focus on learning about a target and gathering information for future attacks. This sort of information gathering can be instrumental to the success of future operations. For example, learning that a target uses a standard IT management software could cause an APT to target this software in a supply-chain vulnerability attack later on. In this lab we'll take a look at some of the social engineering techniques and tools that can help aid Reconnaissance and Resource Development.

Advanced Adversary Tactics - Pivoting with Proxychains
Lab · 1 hour

This lab covers configuring and using proxychains to pivot between machines on a network. The lab simulates pivoting that might occur were a raspberry pi or similar device added to a network.

References

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OWASP Foundation. (2021). OWASP Top Ten Web Application Security Risks. https://owasp.org/www-project-top-ten/